

National Real-Time Payments System — Architecture Brief (UK)

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1. Introduction

This document outlines the architecture and operational design of the National Real-Time Payments System. The intent is to provide a precise, implementation-aware view for an engineering audience, focusing on reliability, performance, security, and settlement integrity.

2. Objectives and Non-Functional Targets

- Instant P2P and P2M payments with end-to-end latency under two seconds.
- Availability target of 99.999% with zero data loss.
- Scale to 150M users, 5M peak concurrent, and 50,000+ TPS.
- Operate under partial bank outages and 10× national spikes.

3. Functional Scope

- Onboarding with KYC; link multiple bank accounts per user.
- Real-time P2P and P2M including QR flows for merchants.
- Transaction history, receipts, and live status (success/pending/failed).
- Bank integration, inter-bank settlement, reconciliation, and chargebacks.
- Merchant registration/verification, dashboards, and daily settlement reports.
- Admin dashboards, regulatory reporting, and audit logs.
- Real-time fraud detection and risk scoring.

4. Architecture Overview

The platform is organized as modular services behind an API gateway. A real-time switching engine orchestrates payment flows via the bank integration layer. Settlement and reconciliation run on scheduled cycles. Fraud detection evaluates transactions in real time without adding user-visible latency. Observability covers metrics, logs, and traces.

5. Security and Compliance

- PCI DSS and ISO 27001 alignment.
- Encryption at rest and in transit.
- Strong authentication and authorization.

6. Reliability and Resilience

- No single point of failure; active-active deployment.
- Automatic failover and graceful degradation.
- Real-time monitoring and alerting.

7. Peak Strategy

- Horizontal scalability for national events (up to 10×).
- Throughput headroom and back-pressure controls.

8. International Readiness

- Currency conversion, cross-border settlement.
- Integration with foreign banking systems and regulations.

9. Closing

The design balances real-time performance with compliance and resilience. Pending alignment on bank connectivity timelines and pilot scope, the platform is positioned for phased rollout and measured scale-up.